Problem sheet: Week 11 - Challenging material

1. You are a statistician who works for a company that carries out political polls. In the run-up to a bi-election you wish to determine the proportion, p, of voters in the constituency who intend to vote for a candidate standing for a certain party (Party X).

   You suspect that people may be reluctant to own up to the intention to vote for this party, so you ask respondents to carry out the following task.

   - Consider the question "Do you intend to vote for Party X?"
   - Roll a fair die once.
   - If a '1' or a '6' is scored give a false answer to the above question.
   - Otherwise give the true answer to the question.

   In a random sample of 500 voters from the constituency, 110 answer 'Yes' to the question.

   Find the MLE of p and identify a 95% CI for p.

2. It is believed that the time to relapse (in years) following treatment for a given condition follows a Gamma(2, λ) distribution where λ is unknown.

   Suppose that for a random sample of 20 patients it is known that 9 of them relapsed within a period of 1 year, but no information on the precise times is available.

   Find the maximum likelihood estimate of λ.

   Hint: You can find considerable help with the solution in the notes.

3. (Harder question.) You are investigating the distribution of the numbers of meteors appearing each night in a certain sector of the sky and you believe this to follow a Poi(λ) distribution where λ is unknown (with numbers being independent over nights).

   On night 1 you count the number of meteors using a perfect detector (detector A) which you have borrowed for one night only, and also using an imperfect detector (detector B) which only detects a given meteor (independently from other meteors) with unknown probability p.

   On night 1 detector A detects 10 meteors and B detects 7 meteors. At the end of night 1 you have to return detector A, leaving you with detector B only. On night 2, you detect 6 meteors with detector B.

   Construct the likelihood $L(\lambda, p)$ for these data and find the MLE of (λ, p).